TripCraft: A Benchmark for Spatio-Temporally Fine-Grained Travel Planning

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Abstract

Travel planning benchmarks for language agents have made real-world planning more concrete, but current evaluations still leave a substantial measurement gap. Existing travel benchmarks are dominated by binary feasibility checks or coarse preference judgments, which are useful for screening invalid plans but weak at distinguishing between itineraries that are all nominally feasible yet differ sharply in schedule realism, spatial coherence, transit accessibility, ordering, and persona fit. We introduce *TripCraft*, a benchmark for persona-conditioned travel planning grounded in a real-world U.S. travel sandbox with flights, accommodations, restaurants, attractions, events, and public-transit metadata. *TripCraft*contains 1,000 queries spanning 140 cities, 3-, 5-, and 7-day trips, and three difficulty levels defined by budget pressure and the number of active local constraints. Each instance is paired with a human-written gold itinerary and rationale produced through multi-round annotation and expert review. Beyond discrete constraint satisfaction, *TripCraft*adds five continuous metrics that score meal timing, attraction pacing, transit proximity, itinerary ordering, and persona alignment. On held-out evaluation splits, parameter-informed prompting improves temporal meal quality substantially, for example from 0.56 to 0.70 on 3-day trips and from 0.61 to 0.80 on 7-day trips, while several discrete pass rates decrease, exposing a previously hidden trade-off between temporal realism and strict feasibility. These results position *TripCraft*as a diagnostic benchmark for studying not only whether an itinerary passes, but how well it is put together.

1 Introduction

Planning is a long-standing testbed for evaluating whether language models can reason over multi-step constraints, maintain consistency, and act on grounded information sources (Valmeekam et al., 2023; Pallagani et al., 2023; Bohnet et al., 2024). Travel planning is especially appealing because it combines symbolic constraints, soft preferences, temporal reasoning, budgeting, and environment grounding in a format that is familiar to end users. Recent benchmarks such as TravelPlanner (Xie et al., 2024), TravelPlanner+ (Singh et al., 2024), and ChinaTravel (Shao et al., 2024) have therefore become useful substrates for evaluating language agents on realistic planning tasks.

However, current evaluation practice remains limited in two ways. First, prior travel-planning evaluations are dominated by pass/fail checks: whether a plan stays within budget, obeys some stated constraints, or is preferred by a judge overall (Xie et al., 2024; Singh et al., 2024; Hao et al., 2025). These criteria are necessary, but they do not distinguish two itineraries that both pass discrete checks while differing markedly in meal timing, visit pacing, public-transit accessibility, or the order in which points of interest (PoIs) are arranged. Second, existing travel tasks typically under-specify transit-aware execution details. In practice, itinerary quality depends not only on what places are selected, but also on when they are visited, how they are chained together, and whether they match the traveler persona under the available city-specific inventory.

*TripCraft*addresses this gap with a benchmark centered on spatio-temporally fine-grained evaluation. The benchmark grounds itinerary generation in a real-world sandbox assembled from flights, accommodations, restaurants, attractions, events, and nearest-stop transit metadata for 140 U.S. cities. Each query contains a natural-language request, structured trip constraints, and a traveler persona. The required output is a day-by-day itinerary with transportation, meals, attractions, accommodations, optional events, and a timestamped PoI sequence. This richer representation supports continuous scoring of itinerary quality in addition to standard feasibility checks.

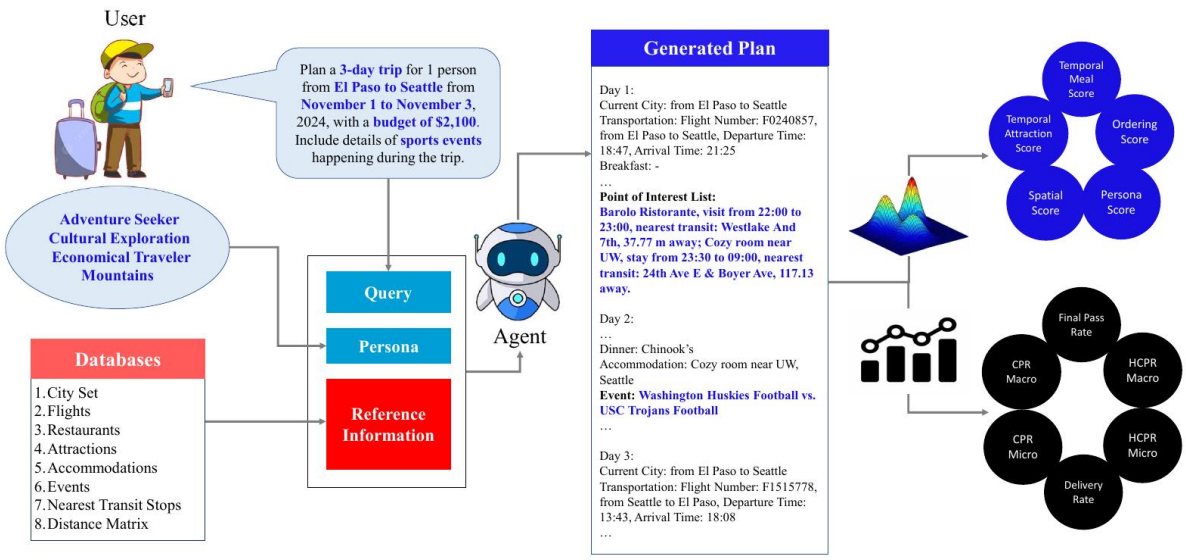

Figure 1: *TripCraft*overview. The benchmark combines a grounded travel sandbox, persona-conditioned itinerary generation, timestamped point-of-interest sequences, and continuous diagnostics for temporal, spatial, sequential, and personalization quality.

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The main contribution of this paper is not a new planning algorithm, but a new evaluation substrate. *TripCraft*contains 1,000 benchmark instances, with 344 3-day trips, 324 5-day trips, and 332 7-day trips. Gold itineraries were written by 25 graduate annotators with multi-round review and final expert adjudication. We use these annotations both as reference itineraries and as empirical sources for parameterizing temporal and attraction-pacing priors. The resulting benchmark enables a different kind of empirical question: not only whether a model can produce a formally valid itinerary, but also which aspects of itinerary quality improve or degrade under different generation strategies.

Using the released implementation, we evaluate GPT-4o under two prompt settings: an original prompt and a parameter-informed prompt that injects annotation-derived priors about meal timing and attraction pacing. The comparison reveals a clear trade-off. Parameter-informed prompting improves temporal meal quality, spatial coherence, ordering, and persona alignment on all trip lengths, but it does not uniformly improve discrete feasibility and in some settings reduces final pass rates. This is precisely the type of mixed, diagnostic signal that a benchmark paper should surface.

In summary, this paper makes four contributions. First, we introduce a grounded travel-planning benchmark with personas, transit metadata, and timestamped itineraries. Second, we define a continuous evaluation suite that measures itinerary quality beyond binary validity. Third, we document the dataset construction, annotation, and validation pipeline used to build *TripCraft*. Fourth, we present empirical findings showing that annotation-informed prompting improves some dimensions of itinerary quality while worsening others, underscoring the need for multi-dimensional evaluation.

2 Related Work

2.1 LLM Planning Benchmarks

Recent work has studied large language models as planners, either in abstract planning environments or in more naturalistic task settings (Valmeekam et al., 2023; Yang and Tomar, 2023; Bohnet et al., 2024; Pallagani et al., 2023). This line of work has been valuable for establishing that apparent fluency does not imply executable planning competence. Benchmarks in this area often emphasize plan validity, task completion, or success under search and verification, which are well matched to classical planning questions (Zhao et al., 2024; Prasad et al., 2024).

Our work is complementary but different in scope. *TripCraft*is not a general planning benchmark; it is a domain-focused evaluation framework for travel planning. The goal is to measure where travel itineraries are realistic, coherent, and personalized, rather than only whether they satisfy an executable end condition.

2.2 Travel Planning Benchmarks and Systems

TravelPlanner introduced a large sandbox and demonstrated that travel planning is a challenging real-world benchmark for language agents (Xie et al., 2024). Follow-up work explored solver-backed verification (Hao et al., 2025), robust planning pipelines (Gundawar et al., 2024), and personalized extensions such as TravelPlanner+ (Singh et al., 2024). TravelAgent similarly studies personalized itinerary construction as an application system (Chen et al., 2024). ChinaTravel extends this direction to Chinese travel planning (Shao et al., 2024).

These benchmarks and systems establish the importance of travel planning, but they leave room for more diagnostic evaluation. TravelPlanner focuses on feasibility in a large sandbox (Xie et al., 2024); TravelPlanner+ adds personalization analysis but relies on preference judgments rather than explicit continuous measurements of temporal and spatial plan quality (Singh et al., 2024). *TripCraft*differs in three ways: it integrates public-transit metadata directly into the itinerary representation, pairs each instance with a human-authored gold itinerary, and adds continuous metrics for schedule realism, ordering, and persona alignment.

2.3 Evaluation Beyond Pass or Fail

Binary pass rates are indispensable for detecting hard violations, but they are blunt instruments for itinerary quality. A plan can satisfy every stated hard constraint and still be awkwardly scheduled, poorly ordered, or mismatched to the traveler persona. *TripCraft*therefore augments feasibility checks with continuous scores. The metric design draws on standard ingredients such as embedding-based semantic similarity using BERT (Devlin et al., 2019) and edit-distance-style sequence comparison (Levenshtein, 1965), but adapts them to travel planning through annotation-derived priors and day-level itinerary structure.

3 Benchmark Design

3.1 Task Formulation

Each *TripCraft*instance asks a model to generate a complete travel itinerary from a natural-language request, structured metadata, and a traveler persona. The input specifies an origin, destination, number of travel days, travel dates, party size, budget, and optional local constraints. The destination is a single city for 3-day trips and a state-level region for 5- and 7-day trips, where the traveler must visit two and three cities respectively. The output is a day-by-day plan containing `current_city`, `transportation`, `breakfast`, `attraction`, `lunch`, `dinner`, `accommodation`, `event`, and a timestamped `point_of_interest_list`.

This representation is richer than a plain list of recommendations. The PoI list explicitly encodes visit order, start and end times, and nearest-transit-stop information. That design makes it possible to evaluate not just content selection but also schedule quality and route structure.

3.2 Grounded Travel Sandbox

Table 1 summarizes the scale of the sandbox. *TripCraft*is built over 140 U.S. cities and includes 3,446,829 flight records, 21,980 events, 5,043 attractions, 3,892 restaurants, 2,800 accommodations, 8,723 nearest transit stops, and a 19,460-entry distance matrix. The benchmark construction described in the code and paper assets uses recent scraped data, OpenStreetMap-style PoI grounding, GTFS-derived public-transit information, and date-aligned event inventories.

Database	Data Entries (#)
City Set	140
Flights	3,446,829
Restaurants	3,892
Attractions	5,043
Accommodations	2,800
Events	21,980
Nearest Transit Stop	8,723
Distance Matrix	19,460

Table 1: Data entries in the database have been scraped to include the most recent data.

The benchmark explicitly incorporates three kinds of structure that are weak or absent in prior travel benchmarks. First, it grounds each PoI in city-consistent inventory, preventing plans from mixing incompatible locations. Second, it attaches nearest-transit-stop information to PoIs, allowing transit accessibility to be evaluated rather than assumed. Third, it supports event and attraction taxonomies that can be used as explicit hard constraints.

3.3 Difficulty Levels and Query Distribution

*TripCraft*contains 1,000 queries spanning three trip lengths and three difficulty levels. The distribution is given in Table 2. The benchmark includes 316 easy, 448 medium, and 236 hard instances overall.

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Days	Easy	Medium	Hard	Total
3-day	130	106	108	344
5-day	87	164	73	324
7-day	99	178	55	332

Table 2: *TripCraft* dataset distribution

The difficulty levels are generated procedurally and differ by the amount of required reasoning. Easy instances have no active local constraints and typically involve a single traveler. Medium instances introduce one local constraint, such as room type, cuisine, event type, or attraction type. Hard instances sample three local constraints from room rules, cuisine requirements, room type, transportation restrictions, event types, and attraction types, while also tightening the effective budget. This construction makes difficulty a property of constraint load and resource pressure rather than simply trip length.

Natural-language queries are then generated from these structured specifications using few-shot prompting with a GPT-4-class model in the released pipeline. Dates are sampled as consecutive days in November 2024, which supports synchronization with the event inventory used during construction.

3.4 Persona and Constraint Design

Each instance includes four persona components: traveler type, purpose of travel, spending preference, and location preference. Table 3 summarizes the benchmark-level constraint vocabulary. Traveler type is particularly important because it conditions attraction-pacing priors used by the continuous metrics.

The benchmark also distinguishes commonsense constraints from hard constraints. Commonsense constraints test general itinerary sanity, such as sufficient meal gaps, non-repeated events, or valid PoI sequences. Hard constraints test satisfaction of explicitly requested requirements, such as room rules, cuisine, transportation restrictions, event types, attraction types, and budget.

3.5 Annotation Protocol

Each benchmark instance is paired with a human-authored gold itinerary and rationale. According to the provided benchmark documentation, 25 graduate annotators produced the itineraries, with approximately 30 minutes spent per instance. Annotation proceeded iteratively: annotators drafted

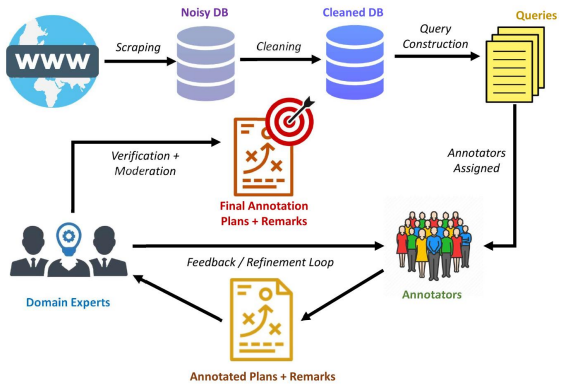

Figure 2: Construction pipeline for *TripCraft*, from grounded data collection and query generation to human itinerary annotation and evaluation parameter estimation.

itineraries, experts provided feedback, and a final review stage combined scripted checks with manual adjudication. We do not report inter-annotator agreement because the benchmark assets do not provide it. Instead, the reliability evidence comes from the combination of structured templates, multi-round review, and the automated validation described in Section 4.

4 Benchmark Validation

Benchmark quality matters as much as benchmark scale. *TripCraft*includes several layers of validation aimed at ensuring that the task is grounded, coherent, and reproducible.

First, the construction pipeline enforces cross-source consistency. Flights, accommodations, restaurants, attractions, and events are all checked against the city sequence in the itinerary. Missing entries are removed or default-filled during data cleaning, and dates are synchronized with the event timeline used in the sandbox.

Second, the released evaluation scripts provide automated sanity checks over generated or annotated plans. Commonsense validation checks that city transitions are reasonable, restaurants and attractions are not repeated, transportation modes are non-conflicting, meal gaps are at least four hours, required fields are present, and the PoI sequence begins and ends with the appropriate accommodation. Sandbox validation further verifies that cited flights, restaurants, attractions, events, accommodations, and nearest-transit-stop distances exist in the underlying database.

Third, *TripCraft*uses annotation statistics to calibrate continuous metrics instead of hand-setting all

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Commonsense Constraints	
Sufficient Meal Gaps	Defines a minimum gap of four hours between the meals.
Valid PoI list	Defines validity rules for the Point of Interest list.
Diverse Events	Event choices should not be repeated throughout the trip.
Hard Constraints	
Event Types	Event Types include four distinct categories—Sports, Arts & Theatre, Music, and Film.
Attraction Types	Each attraction belongs to one or more of 15 predefined categories, ensuring a well-distributed selection of activities.
Persona Components	
Traveler Type	Defines how a traveler approaches their journey—whether they seek relaxation in cozy spots or adrenaline-pumping adventures.
Purpose of Travel	Captures trip motivation. Examples: to unwind, explore cultures etc.
Spending Preference	Reflects the traveler's budget and style, from luxurious indulgence to cost-conscious experiences.
Location Preference	Highlights preferred environments, such as beaches, mountains, cities, or wildlife-rich forests.

Table 3: Constraint families and persona components used in *TripCraft*. These choices reflect the coverage and limitations of the grounded travel sandbox.

temporal priors. Table 4 shows representative day-wise parameters estimated from the annotation distribution. These parameters include meal duration statistics and traveler-type-specific attraction-count priors. This design grounds the continuous metrics in observed human planning behavior rather than arbitrary constants.

These validation steps do not eliminate all benchmark risk. In particular, persona quality still depends on the expressiveness of the textual persona fields, and expert review does not substitute for formal inter-annotator agreement. We return to these limitations in Section 8.

Parameter	**3-day**	**5-day**	**7-day**
Restaurant Parameters			
Breakfast			
Mean Duration (hrs)	0.90	1.08	0.85
Std. Duration (hrs)	0.24	1.43	0.23
Lunch			
Mean Duration (hrs)	1.11	1.10	0.99
Std. Duration (hrs)	0.36	0.35	0.26
Dinner			
Mean Duration (hrs)	1.19	1.32	1.15
Std. Duration (hrs)	0.43	0.91	1.15
Attraction Parameters			
$\lambda_{laidback}$	1.10	1.26	1.11
$\lambda_{adventurous}$	2.01	1.61	1.82
σ_d (hrs)	1.11	1.07	0.90
n^{max}	5	4	4
n^{min}	0	0	0
k (hrs)	0.28	0.28	0.28

Table 4: Day-wise parameter details calculated from our annotation distribution. We report representative meal and attraction-pacing parameters used by the continuous metrics.

5 Evaluation Protocol

5.1 Discrete Feasibility Metrics

*TripCraft*retains discrete evaluation because hard failures matter. We report delivery rate, commonsense pass rate (CPR), hard-constraint pass rate (HCPR), and final pass rate. The micro versions aggregate correctness across individual checks, while the macro versions count a plan as correct only when it satisfies the full group of relevant constraints. Final pass rate requires both commonsense and hard constraints to pass jointly.

These metrics are intentionally strict. They surface brittle failures such as invalid city references, missing transportation, budget violations, repeated attractions, incorrect room types, or unsupported event types.

5.2 Continuous Quality Metrics

The main novelty of *TripCraft*is a suite of five continuous metrics that score complementary aspects of itinerary quality.

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Temporal Meal Score. For breakfast, lunch, and dinner, we score the midpoint and duration of each meal against annotation-derived timing distributions. Higher values indicate meal scheduling that more closely matches the empirical timing patterns in the gold itineraries.

Temporal Attraction Score. This score measures whether attraction visit durations and the number of attractions per day match the observed pacing of the corresponding traveler type. The implementation combines attraction-category duration priors with a Poisson prior over daily attraction count, conditioned on whether the persona is laid-back or adventurous.

Spatial Score. The spatial metric rewards plans whose PoIs are close to their nearest transit stop. The score is computed from the nearest-stop distances included in the PoI list, using a decay function that assigns higher values to more transit-accessible itineraries.

Ordering Score. This metric measures how well the generated itinerary order matches the annotated reference plan. The implementation converts each day into a type sequence over accommodations, restaurants, and attractions, then applies a weighted edit-distance similarity (Levenshtein, 1965).

Persona Score. The persona metric estimates how well selected PoIs align semantically with the stated traveler persona. The released implementation embeds the persona fields and PoI names with BERT (Devlin et al., 2019) and averages cosine similarity across persona components and visited places.

Together, these metrics make the benchmark diagnostic. They allow two plans with the same final pass/fail outcome to receive meaningfully different quality scores.

6 Experimental Setup

The released code evaluates a single planning backbone, GPT-4o, under two prompt settings. The *w/o Parameter Info* setting uses an original direct planning prompt. The *w/ Parameter Info* setting augments the prompt with annotation-derived guidance about preferred meal times, meal durations, and attraction pacing. The prompt text in the release makes this intervention explicit, for example by suggesting ideal breakfast, lunch, and dinner times and contrasting laidback versus adventure-seeking behavior.

Generated itineraries are post-processed into structured JSON using a GPT-4-based conversion pipeline adapted from TravelPlanner (Xie et al., 2024). The code README reports that all experiments were run on a single NVIDIA L40 GPU setup. We evaluate on the held-out subsets used by the released scripts: 230 3-day instances, 230 5-day instances, and 227 7-day instances. For the continuous metrics, annotated itineraries serve as references for ordering and as the empirical basis for the temporal parameterization.

This experimental setup should be read as a benchmark demonstration rather than a model leaderboard. The purpose is to show what *TripCraft*reveals when the same planning model is evaluated under two different prompting conditions.

7 Results and Analysis

7.1 Continuous Metrics Reveal Quality Differences

Table 5 reports the five continuous metrics. Parameter-informed prompting improves temporal meal quality for all trip lengths, from 0.56 to 0.70 on 3-day trips, from 0.69 to 0.75 on 5-day trips, and from 0.61 to 0.80 on 7-day trips. Spatial and ordering scores also improve consistently, though more modestly: for example, 3-day spatial score rises from 0.80 to 0.83 and ordering score from 0.70 to 0.74, while 7-day ordering improves from 0.96 to 0.97.

Persona alignment increases slightly in all settings, but the gains are small in magnitude, rising from 0.46 to 0.48 for 3-day trips and from 0.50 to 0.51 for 7-day trips. Temporal attraction quality is mixed. It improves for 3-day and 5-day trips, but decreases on 7-day trips from 0.0122 to 0.0057. The small absolute scale of this metric reflects its probability-based construction, so the relevant comparison is within metric rather than across metrics.

These results support two conclusions. First, annotation-derived parameters help the model produce schedules that look more human-like in time and order. Second, different aspects of itinerary quality do not move together. In particular, attraction pacing and persona fit remain comparatively difficult even when meal timing improves substantially.

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Setting	Category	$\bar{T}_{meal}$	$\bar{T}_{attrac}$	$\bar{S}_{spatial}$	$\bar{S}_{persona}$	$\bar{S}_{ord}$
w/o Parameter Info	3-day	0.56	0.0169	0.80	0.46	0.70
	5-day	0.69	0.0057	0.86	0.49	0.91
	7-day	0.61	**0.0122**	0.85	0.50	0.96
w/ Parameter Info	3-day	**0.70**	**0.0171**	**0.83**	**0.48**	**0.74**
	5-day	**0.75**	**0.0078**	**0.87**	**0.50**	**0.93**
	7-day	**0.80**	0.0057	**0.86**	**0.51**	**0.97**

Table 5: Results of our proposed evaluation metrics for the two settings. For each category, the **bolded** values represent the better ones between the two settings.

7.2 Discrete Metrics Expose a Feasibility Trade-off

Table 6 reports the existing discrete metrics. The picture here is mixed. On 3-day trips, parameter-informed prompting improves delivery rate from 92.60% to 96.08% and CPR micro from 77.61% to 80.08%, but lowers CPR macro from 6.08% to 5.21%, HCPR macro from 26.08% to 25.21%, and final pass rate from 3.47% to 1.74%.

The trade-off is more pronounced on 5-day trips. Delivery rate falls from 98.69% to 91.30%, CPR micro from 66.99% to 61.69%, CPR macro from 2.17% to 0.87%, and final pass rate from 1.74% to 0.43%, even though HCPR micro improves from 4.21% to 8.60%. For 7-day trips, final pass rate is 0.00% in both settings, while HCPR micro improves slightly from 0.62% to 0.83%.

The key lesson is that stronger temporal priors do not automatically translate into fully feasible plans. In some cases they appear to make the generator more rigid, which can improve schedule regularity while hurting constraint satisfaction or end-to-end executability.

7.3 Why Continuous Evaluation Matters

Figure 3 illustrates the central motivation for *TripCraft*. Constraint-based metrics can rate two plans similarly even when one is clearly better organized for actual use. The continuous metrics differentiate these cases by assigning separate scores for temporal realism, spatial coherence, order quality, and persona alignment.

This diagnostic behavior is visible in the aggregate results as well. On 7-day trips, ordering scores are already very high in both settings, suggesting that itinerary skeletons are relatively easy to imitate from the prompt format. Yet final pass rate remains zero, which indicates that long-horizon feasibility is bottlenecked by other issues such as precise constraint satisfaction, inventory grounding, or budget consistency. Conversely, the large improvements

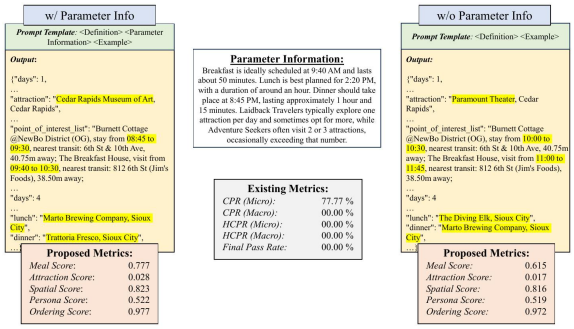

Figure 3: Why continuous evaluation matters. Existing constraint-based metrics can rate two plans equally even when they differ in temporal, spatial, and persona quality; *TripCraft*assigns graded scores to these dimensions.

in meal timing would be invisible under a purely binary evaluation.

8 Discussion

*TripCraft*contributes a more informative way to evaluate travel planning systems. It measures whether generated itineraries are not only constraint-aware but also temporally plausible, transit-aware, reasonably ordered, and at least partially aligned with traveler personas. In this sense, the benchmark is best viewed as an evaluation protocol for fine-grained itinerary quality, not merely as a harder set of travel queries.

At the same time, *TripCraft*has clear limitations. The benchmark is restricted to U.S. cities and to the data snapshot used during construction. Because dates are sampled from a bounded period, event coverage and seasonality are only partially represented. Persona alignment is measured through embedding similarity over textual persona fields and PoI names, which is an imperfect proxy for user satisfaction. The benchmark assets describe multi-round annotation and expert adjudication, but they do not report inter-annotator agreement. Finally, the experiments in this paper compare only

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Setting	Category	Delivery Rate	CPR		HCPR		Final Pass Rate
			Micro	Macro	Micro	Macro	
w/o Parameter Info	3-day	92.60	77.61	**6.08**	**27.63**	**26.08**	**3.47**
	5-day	**98.69**	**66.99**	**2.17**	4.21	3.04	**1.74**
	7-day	**96.03**	**70.35**	0.00	0.62	0.00	0.00
w/ Parameter Info	3-day	**96.08**	**80.08**	5.21	26.10	25.21	1.74
	5-day	91.30	61.69	0.87	**8.60**	**5.21**	0.43
	7-day	92.51	66.43	0.00	**0.83**	0.00	0.00

Table 6: Results of existing evaluation metrics for the two settings. CPR and HCPR stand for Commonsense Pass Rate and Hard Constraint Pass Rate respectively. For each category, the **bolded** values represent the better ones between the two settings. Moreover, equal values have been underlined.

two prompt settings of a single planner backbone, so the current empirical section should be read as a case study of what the benchmark can reveal, not as a comprehensive model ranking.

These limitations do not reduce the value of the benchmark; they define its intended use. *TripCraft*is most useful when researchers want to understand which aspects of itinerary generation improve under a new model, prompting strategy, planner, or verifier, and which aspects remain brittle.

9 Conclusion

We presented *TripCraft*, a benchmark for spatio-temporally fine-grained travel planning. The benchmark combines a grounded travel sandbox over 140 U.S. cities with persona-conditioned queries, human-authored gold itineraries, and an evaluation protocol that mixes strict feasibility checks with continuous quality scores. Experiments with GPT-4o under two prompt settings show that annotation-derived parameter information substantially improves temporal meal quality and modestly improves spatial, ordering, and persona scores, while not consistently improving discrete feasibility. This mixed outcome is exactly the point of the benchmark: *TripCraft*enables researchers to see where itinerary generation succeeds, where it fails, and which trade-offs are hidden by binary evaluation alone.

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